

ProcurePing — Full PRD + System Design

(Founders/Engineering Spec v1)

Scope: cross-industry reverse-search e-commerce where **buyers post an intent with budget** → **verified suppliers receive pings** → **apples-to-apples offers** → **escrowed checkout** → **fulfillment, dispute, payout**. Built to start with construction, but generalized for *any* tradable SKU/service.

1) Product Goals & Non-Goals

Goals - Cut buyer procurement time from hours/days to minutes by broadcasting structured requests (Intents) to qualified suppliers. - Guarantee serious demand with **pre-authorized funds/escrow**. - Normalize offers for true comparisons; detect scammy pricing via **price-band intelligence**. - Provide end-to-end transaction safety: KYC/KYB, milestone tracking, dispute resolution, and compliant payouts. - Multi-category, multi-geography, multi-currency from day 1. - **Zero take-rate**: No markup, no commissions on transactions. Platform revenue is **ads/promotions only** (clearly labeled; no ranking bias without disclosure).

Non-Goals (MVP) - No public browsing marketplace (traditional catalog) beyond supplier profiles; the core flow is Intent → Offers. (Catalog browsing can be Phase 2.) - No custom logistics network—use 3rd-party shippers first (Grab/Lalamove, LBC, NinjaVan, etc.). - No on-platform lending at launch (only payments/escrow). Financing partners in Phase 3.

Primary KPIs - Time-to-First-Offer (TTFO), Offers per Intent, Award Rate, On-Time Delivery %, Dispute Rate, GMV, % of Intents funded, Ad CTR/eCPM, Net Ad Revenue.

2) Core Concepts & Entities

- **User**: buyer, supplier agent, supplier admin, platform staff.
- **Company**: Supplier business with one or more Branches.
- **Branch**: physical location with delivery radius and inventory.
- **Category**: industry vertical (Construction, Marine, Auto, IT, etc.).
- **Attribute Schema**: normalized key/value pairs for each Category (e.g., Rebar: size, grade, length; Paint: brand, line, color code, sheen; Wires: AWG, strands, insulation type).
- **CatalogItem (SKU Template)**: a templated item spec, optional GTIN/UPC/MPN; suppliers map their SKUs to templates.
- **Intent**: buyer request: specs (attributes), qty + unit, budget range, delivery window, location, notes, attachments.
- **Offer**: supplier response to an Intent (unit price, availability, ETA, delivery fees, warranty, total landed cost, tier).
- **Order**: created when buyer awards an Offer.

- **EscrowTransaction:** auth, capture, release, refund, chargeback links.
 - **Delivery:** fulfillment milestones + proofs.
 - **Review/Rating:** post-order evaluation.
 - **Dispute:** evidence threads + resolutions.
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3) User Roles & Permissions (RBAC)

- **Buyer:** create/manage Intents, award Offers, pay, receive goods, open disputes, review.
 - **Supplier Admin:** company verification, manage branches, users, catalogs, pricing, offer rules, banking.
 - **Supplier Agent:** receive pings, submit Offers, chat, manage orders.
 - **Platform Ops:** KYC/KYB reviews, dispute mediation, refunds, fraud actions, configurations.
 - **Auditor (read-only):** access to logs/reports.
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4) End-to-End Flows

4.1 Buyer Flow

1. **Create Intent** (web/mobile):
2. Choose Category → dynamic Attribute Schema form renders.
3. **Search modes:**
 - **Spec Form** (structured fields),
 - **Natural Language** (NLP parses attributes),
 - **Image Search** (upload photo; system suggests likely categories/specs using vision model; buyer confirms).
4. Upload images/docs (e.g., plan detail, shelf tag).
5. Enter qty + unit; delivery window; location pin + radius; **budget min/max**.
6. Choose payment method; toggle **Pre-Authorize Funds** (deposit% or full budget).
7. Submit → Intent ID created → PSP auth hold initiated → success = Intent set to **ACTIVE**.
8. **Live Offers**
9. Offers stream in (push/websockets) with countdown timers.
10. **Offer Matrix:** columns for total price, unit price, ETA, ratings, distance, warranty, stock confidence.
11. **Price-Band Ribbon** visible with P10–P90 regional band; outliers flagged.
12. **Award**
13. Buyer selects Offer → Order created; **capture** selected amount; release other holds.
14. **Fulfillment**
15. Milestones: **READY_FOR_PICKUP** / **DISPATCHED** / **DELIVERED** / **ACCEPTED**.
16. Buyer taps **Accept** to release escrow; or opens **Dispute**.

4.2 Supplier Flow

1. **Onboarding:** KYB (company docs), TIN/BIR/SEC/DTI uploads, bank account verification, categories, coverage radius per branch.

2. **Catalog Mapping:** bulk CSV import or API; map to Category Attribute Schemas; set matching tags; optional **Alert Subscriptions** and **Auto-Offer rules** (e.g., auto-reply with fixed unit price for intents within price/qty bounds).
3. **Receive Ping:** real-time notification; open Intent; one-click **Make Offer**.
4. **Offer Composer:** unit price, stock availability (firm/backorder with date), delivery fee & ETA, min/max qty, warranty, tier (good/better/best), Offer expiry.
5. **Order Fulfillment:** accept order; upload proof; request payout on delivery.

4.3 Payments & Escrow States

- AUTH_PENDING → AUTH_HELD → (on award) CAPTURED → (on acceptance) RELEASED → (refund/partial) REFUNDED.
- Dispute path places automatic hold and pauses auto-release.

5) Matching Logic (MVP to ML)

Supplier Notification Preferences - Each supplier/branch can configure **Alert Subscriptions**: - Attribute watchlists (e.g., brand=Holcim, gauge=26, color=Seafoam). - **Spec templates** (JSON) to match ranges (e.g., price 450–520, qty $\geq$ 100, radius $\leq$ 30km). - **Creative/Custom work** opt-in (e.g., dressmaker: category=Apparel, type="dress", allow text-only/image-only intents with freeform quoting). - Notification channels: push/SMS/email; quiet hours; rate limits.

Phase 1 — Rules - Supplier eligibility: Category match + required attributes + geo radius + verification level $\geq$ Basic + supplier status = ACTIVE + subscription match. - Score = Attribute Match (0–50) + Proximity (0–20) + Reliability (0–20) + Subscription Specificity (0–10). - Notify suppliers with score $\geq$ threshold (e.g., 65). Cap invites (e.g., top 50) to avoid spam.

Phase 2 — Heuristics - Learn acceptance probability per supplier/SKU/geo/time; boost reliable fast responders. - Price-band model: rolling window by Category + AttrHash + Region → compute P10, P50, P90; flag quotes < P10 (risk) or > P90 (overpriced).

Phase 3 — ML - Offer ranking using gradient models (features: price, ETA, reliability, historical buyer preferences, distance, stock confidence). A/B tested.

6) Fraud, Safety & Compliance

- **KYB/KYC:** Verify identity, corporate existence, tax info, beneficial owners (where required).
- **Payment Risk:** 3DS/SCA where available; device fingerprint; velocity checks.
- **Counterfeit/Quality:**
 - Price outlier flags; required **Brand Authenticity Docs** upload for flagged items.
 - Counterfeit checklist per category (e.g., cement bag print markers, rebar mill certs, paint batch codes).
- **Holdbacks:** new suppliers have 3% payout reserve for 90 days; manual review for disputes.
- **Privacy:** mask buyer details pre-award; secure messaging.

- **Compliance:** data protection (GDPR/PDPA), marketplace terms, escrow agreement, e-invoicing, VAT handling.
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7) Monetization

- **No commissions. No markups.** Transactions are fee-free from ProcurePing.
- **Revenue = Advertising & Promotions only**, and always clearly labeled:
- Sponsored supplier profiles, sponsored categories, and **Promoted Offers** (never auto-ranked above organic without "Sponsored" label).
- **New Product Launch units** (carousel, inbox announcements) purchased by suppliers/brands.
- **Contextual ads** in non-decision surfaces (dashboards, learning center). No ads inside the award/checkout step.
- **Fees pass-through:** PSP/escrow fees are shown transparently and charged at cost to buyer (default) or supplier (configurable per order). Platform does not add margin.
- **Optional Paid Add-ons (non-transactional):** verification fast-track, analytics pack, API access, extra alert seats. (All optional and not required to win deals.)

Ranking Integrity Policy - Organic ranking based on Match/Trust/Price/ETA only. Ads do **not** influence organic rank; sponsored units live in clearly separated placements.

8) System Architecture (Proposed)

Overall: Modular services with event bus. - **Frontend:** React/Next.js (web), React Native (iOS/Android). Websockets for live offers. - **Search/Understanding:** - **Vision service** (image → embedding + candidate categories/specs) using on-device or server model; stores embeddings in vector DB (FAISS/PGVector). - **NLP extraction** (text intent → attributes) using rule-based + ML; confidence scores with human edit. - **Backend:** Node.js (NestJS/Express) or Python (FastAPI). Service modules: - Auth & RBAC - Catalog & Attribute Schemas - **Understanding (NLP/Vision)** - Intent Service - Matching & Notifications - Offer Service - Orders & Escrow - Delivery & Proofs - Reviews & Trust Graph - Disputes - Admin/Ops - **Datastores:** - PostgreSQL (core relational data, JSONB for attributes) - Redis (caching, sessions, rate limiting) - Elasticsearch/OpenSearch (search & analytics) - Vector DB (PGVector/FAISS) for image/spec embeddings - S3/GCS (attachments & proofs) - **Messaging/Event Bus:** Kafka or RabbitMQ (events: IntentCreated, OfferSubmitted, OrderAwarded, DeliveryUpdated, DisputeOpened). - **Payments/PSP:** Stripe (global) + regional options (Xendit/PayMongo). Webhooks for auth/capture/refund events. PSP fees passed through at cost. - **Geo:** PostGIS for distance queries. - **Notifications:** OneSignal/FCM/APNs, email (Sendgrid), SMS (Twilio/Infobip).

SLOs - 99.9% uptime for core APIs. - P95 TTFO push < 5s from Intent creation to first supplier notification.

9) API Surface (MVP)

Example endpoints (JSON). Auth: OAuth 2.0 / JWT. All timestamps ISO-8601 UTC.

Auth - POST /auth/register (role, email, phone, password) - POST /auth/login - POST /auth/verify (KYC/KYB uploads)

Catalog & Schemas - GET /categories - GET /categories/{id}/schema - POST /supplier/catalog/import (CSV) - POST /supplier/catalog/item {attrs, price, stock}

Intent - POST /intents {category_id, attrs, qty, unit, budget_min, budget_max, location{lat,lng,radius_km}, delivery_window{start,end}, notes, preauth:true} - Response: {intent_id, escrow:{status, auth_amount}} - GET /intents/{id} - GET /buyer/intents?status= - POST /intents/{id}/cancel

Offers - POST /intents/{id}/offers {unit_price, qty_available, delivery_fee, eta_date, warranty, tier, expires_at, stock_confidence} - GET /intents/{id}/offers

Award & Orders - POST /offers/{id}/award → creates Order, captures funds. - GET /orders/{id} - POST /orders/{id}/milestones {status, notes, photos[]} - POST /orders/{id}/accept → release escrow - POST /orders/{id}/dispute {reason, evidence[]}

Messaging - POST /intents/{id}/messages - GET /intents/{id}/messages

Reviews - POST /orders/{id}/review {rating, text}

Admin/Ops - GET /admin/verification/queue - POST /admin/disputes/{id}/resolve {decision, refund_amount}

Webhooks - /webhooks/payments (auth/capture/refund/chargeback) - /webhooks/notifications (delivery receipts)

10) Database Sketch (PostgreSQL)

users (id, role, email, phone, password_hash, status, created_at)

companies (id, name, tax_id, country, verification_level, status)

branches (id, company_id, name, lat, lng, radius_km, hours, delivery_methods)

categories (id, name, slug, schema_json)

catalog_items (id, company_id, branch_id, category_id, attrs_jsonb, price, stock_qty, status, tags[])

intents (id, buyer_id, category_id, attrs_jsonb, qty, unit, budget_min, budget_max, lat, lng, radius_km, window_start, window_end, notes, status)

offers (id, intent_id, company_id, branch_id, unit_price, qty_available, delivery_fee, eta_date, warranty, tier, stock_confidence, expires_at, status, total_price_cached)

orders (id, offer_id, buyer_id, company_id, total_amount, currency, status)

escrow (id, order_id, psp, auth_amount, captured_amount, released_amount, status)

deliveries (id, order_id, status, event_time, actor_id, notes, proofs[])

messages (id, thread_type, thread_id, sender_id, body, attachments[], created_at)

reviews (id, order_id, rater_role, rating, text)

disputes (id, order_id, reason, status, resolution, refund_amount)

supplier_stats (company_id, on_time_pct, cancel_rate, dispute_rate, response_time_ms, rating_avg)

Indexes: GIN on attrs_jsonb & tags, GiST on (lat,lng) via PostGIS, composite (intent_id, status), etc.

11) State Machines

Intent: DRAFT → ACTIVE → EXPIRED | CANCELED | AWARDED

Offer: SUBMITTED → WITHDRAWN | EXPIRED | AWARDED | REJECTED

Order: CREATED → IN_PROGRESS → DELIVERED → ACCEPTED → PAID | DISPUTED → RESOLVED

Escrow: AUTH_PENDING → AUTH_HELD → CAPTURED → RELEASED | REFUNDED | CHARGEBACK

12) Price-Band & Risk Scoring

- Data inputs: recent Offers & Orders by `AttrHash + Region + Currency` (e.g., 60-day rolling). Remove extreme outliers (MAD filtering). Compute P10/P50/P90.
 - UI: show band and median; label quotes <P10 as **risk** (tooltip: potential counterfeit, short-weight, bait). >P90 flagged as **premium**.
 - Supplier Risk Score = $w_1 on_time + w_2 dispute_rate + w_3 response_speed + w_4 rating + w_5 tenure$.
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13) Notifications & Templates

- **Supplier Ping (push/SMS/email):** "New pre-funded request in *[Category]*: *[Qty & Item]*, budget *[range]*, ETA *[date]*. Reply in *[timer]*."

- **Expiring Offer Reminder:** "Your offer on Intent #123 expires in 30m. Edit or extend now."
 - **Award Notice:** "Congrats! Intent #123 awarded. Funds captured. Prepare for delivery by *[date]*."
 - **Delivery Acceptance:** "Buyer accepted delivery. Payout T+0/1."
 - **Dispute Opened:** "Order #456 is disputed. Provide evidence by *[deadline]*."
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14) Admin Console (Ops)

- Verification queue (document viewer, checklist, approve/deny, risk notes).
 - Live Intents dashboard (age, offers count, TTFO, geographic heat map).
 - Dispute center (timeline, evidence, guided resolution, partial refunds).
 - Pricing intelligence (bands per SKU/region; anomaly detector).
 - Audit logs (who changed what/when).
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15) Security & Privacy

- JWT with short TTL + refresh tokens; per-role scopes.
 - 2FA (TOTP/SMS). Passwordless (email link) option.
 - PII encryption at rest (KMS). S3 signed URLs for files.
 - Rate limiting, WAF, IP allow/deny, audit trails.
 - GDPR/PDPA tools: data export, delete, consent logs.
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16) Internationalization & Tax

- i18n from the start (copy keys).
 - Multi-currency via PSP and FX table; store monetary amounts in minor units (cents).
 - Tax/VAT: store tax profile per region; line-item tax rates in orders and receipts.
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17) Analytics

- Event tracking (Segment) for: IntentCreated, OfferSubmitted, OfferViewed, OfferAwarded, OrderDelivered, DisputeOpened.
 - Funnel dashboards per category/geo.
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18) QA & Testing

- Unit tests $\geq$ 80% critical modules (Matching, Escrow, Orders).
- Contract tests for webhooks.
- Load tests: 10k concurrent supplier pings, <2s P95.
- UAT scripts per role; sandbox PSP keys.

19) Rollout Plan

- **Alpha:** Cebu (Construction). 50 suppliers across 10 SKUs. Concierge support.
 - **Beta:** Add Marine/Auto categories. Expand to Metro Manila + Gainesville, GA.
 - **GA:** Open self-serve onboarding, add financing partners, push mobile apps.
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20) Open Design Decisions

- Deposit % default (20% vs full budget) per category risk.
 - Auto-award to best-ranked offer at timer expiry? (Off by default.)
 - Split awards (multi-supplier) for large qty—Phase 2.
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21) Sample JSON Payloads

Create Intent (Buyer)

```
{
  "category_id": "construction.cement",
  "attrs": {"brand": "Holcim", "type": "Portland", "bag_weight_kg": 40},
  "qty": 500,
  "unit": "bag",
  "budget_min": 240000,
  "budget_max": 260000,
  "currency": "PHP",
  "location": {"lat": 10.3157, "lng": 123.8854, "radius_km": 30},
  "delivery_window": {"start": "2025-10-25T08:00:00Z", "end":
    "2025-10-26T17:00:00Z"},
  "notes": "Need factory fresh, no clumped bags.",
  "preauth": true
}
```

Offer (Supplier)

```
{
  "unit_price": 495,
  "qty_available": 500,
  "delivery_fee": 5000,
  "eta_date": "2025-10-25",
  "warranty": "Replacement for damaged bags on delivery",
  "tier": "good",
}
```

```
"stock_confidence": "firm",
"expires_at": "2025-10-24T10:00:00Z"
}
```

22) Appendix A — Category Attribute Schemas (starter)

Construction: Cement, Rebar, CHB, Sand/Gravel, Plywood, Paint, Roofing, Wires, PPR/PVC, Fasteners, Waterproofing. **Marine:** Gelcoat, Fiberglass mat (oz), Props (diameter/pitch), Batteries (type/Ah/CCA), Bilge pumps (GPH), Chartplotters (screen size/maps). **Auto:** Tires (size/speed/load), Oils (grade/viscosity), Brake pads (model fitment), Batteries (group/CCA), Filters (part #). **IT/Office:** Laptops (CPU/RAM/SSD), Switches (ports/PoE), Printers (type/ppm), Chairs/Tables (dims/materials).

Each schema defines: required fields, value types, acceptable ranges, and units.

23) Appendix B — UI Copy & UX Rules

- Replace “Start Search” with “**Post Request**”.
 - Show running **offer count** and **time left** in the Intent header.
 - Comparison table default sort = **Total Landed Cost**; secondary sort = **ETA**.
 - Always render **Price-Band** component when ≥ 5 comparable data points exist.
 - Display **Supplier Trust Badge** levels with hover details (docs verified, on-time %, dispute rate).
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24) Appendix C — SLA & Policies (Buyer/Supplier)

- **Supplier response SLA:** target ≤ 30 min; slow responders suppressed in ranking.
 - **Cancellation:** supplier cancels post-award $\rightarrow$ strike and fee; buyer cancels post-award $\rightarrow$ partial fee unless non-fulfillment.
 - **Dispute timelines:** evidence within 72h; resolution within 7 days unless escalated.
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25) Roadmap Summary

- **Phase 0 (0–90d):** Concierge MVP, 2 categories, Stripe/Xendit, Cebu pilot, web only.
- **Phase 1 (3–6m):** Full portals, escrow automation, reviews, price-bands, mobile apps beta.
- **Phase 2 (6–12m):** ML ranking, financing partners, multi-region tax, catalog browsing, split orders, enterprise APIs.